



Shortest Path, Homework 10

Algorithm 1: two_Mixed_Sorting($A[1, \dots, n]$)

```
1  $L_1, L_2 \leftarrow$  two empty linked lists
2  $L_1.append(A[1])$ 
3 for  $i \in [2, \dots, n]$  do
4   if  $|A[i] - A[1]| \leq \frac{n}{k}$  then
5      $L_1.append(A[i])$ 
6   else
7      $L_2.append(A[i])$ 
8 Sort  $L_1$  and  $L_2$  using counting sort.
9 if  $L_1.first < L_2.first$  then
10   Output  $L_1, L_2$ 
11 else
12   Output  $L_2, L_1$ 
```

Algorithm 2: $\text{kMixedInsert}(\text{node}, x)$

```

1 if  $|x - \text{node}.x| \leq \alpha$  then
2    $\text{node.list.append}(x)$ 
3 else if  $x < \text{node}.x$  then
4   if  $\text{node.left is Null}$  then
5      $\text{node.left} = \text{new Node}(\text{parent} \leftarrow \text{node}, x \leftarrow x, \text{list} \leftarrow \text{new linked-list}(\{x\}), \text{height} \leftarrow 0)$ 
6   else
7      $\text{kMixedInsert}(\text{node.left}, x)$ 
13 else if  $x > \text{node}.x$  then
14   if  $\text{node.right is Null}$  then
15      $\text{node.right} = \text{new Node}(\text{parent} \leftarrow \text{node}, x \leftarrow x, \text{list} \leftarrow \text{new linked-list}(\{x\}), \text{height} \leftarrow 0)$ 
16   else
17      $\text{kMixedInsert}(\text{node.right}, x)$ 
23 Recompute the height of the node.
24 If needed, apply a rotation to rebalance the subtree.

```

Algorithm 3: $\text{k-Mixed-Sorting}(A[1, \dots, n], k)$

```

1  $\alpha \leftarrow \lceil \frac{n}{k} \rceil$ 
2  $T \leftarrow \text{empty Avl}()$ 
3 for  $i \in [1, \dots, n]$  do
4    $\text{kMixedInsert}(T.\text{root}, A[i])$ 
5 end
6  $B \leftarrow \text{output the lists in the } T \text{ according to 'Inorder'-order.}$ 
7 for  $i \in [1, \dots, k]$  do
8    $\text{Sort } B[i] \text{ using counting sort.}$ 
9    $\text{output } B[i] \text{'s elements in order.}$ 
10 end

```

Algorithm 4: Sort^{*}($A[1, \dots, n], k$)

```

1  $H \leftarrow \text{Build MinHeap}(A[1], A[2], \dots, A[k + 1])$ 
2  $B \leftarrow$  new empty  $n$ -length array.
3 for  $i \in [1, \dots, n]$  do
4    $B[i] \leftarrow \text{ExtractMin}(H)$ 
5   if  $i + k + 1 \leq n$  then
6      $\text{Insert}(H, A[i + k + 1])$ 
7   end
8 end
9 return  $B$ 

```

Algorithm 5: Sort_one($A[1, \dots, n]$)

```

1 for  $i \in [1, \dots, n]$  do
2   if  $A[i] > A[i + 1]$  then
3      $\text{Swap } A[i], A[i + 1].$ 
4   end
5 end
6 return  $A$ 

```
